

Conclusion

The detection of primordial B-mode polarisation in the Cosmic Microwave Background would provide direct evidence for Inflation, yet it remains one of the most demanding measurements in observational cosmology: the expected signal is extremely faint, and it is buried under astrophysical foregrounds, gravitational lensing, and, for ground-based experiments, atmospheric emission. QUBIC addresses this challenge with an instrumental strategy that differs from that of conventional imagers, combining bolometric interferometry, self-calibration, and spectral imaging to control systematic effects and to separate the primordial signal from its contaminants. This thesis has followed that strategy from the instrument itself to the reconstruction algorithms it enables, and finally to an innovative application of spectral imaging to atmospheric mitigation.

The instrument and its calibration. Chapter 2 details the specific instrumental design of the QUBIC instrument, explaining the different optical elements and its scientific goal. After, Chapter 3 presented the self-calibration procedure and examined the noise of the GPS system associated with QUBIC's calibration source, propagating its statistical properties through to the position and orientation of the source and, from there, to the amplitude of its beam. The resulting uncertainty was found to be small, suggesting that GPS noise alone should not be a limiting factor for self-calibration, though this conclusion should be revisited once on-site operational data are available. Chapter 4 then formalised the spectral imaging principle exploited throughout the rest of the manuscript: the frequency dependence of QUBIC's synthesized beam allows the reconstruction of several sub-bands within a single physical band, at the cost of increasingly anti-correlated noise between them, to improve the instrument spectral resolution.

Map-making, component separation, and simulation software. The central part of this thesis developed two complementary answers to the reconstruction problem using spectral imaging. Frequency Map-Making (Chapter 5) reconstructs several frequency maps within the physical bandwidth of QUBIC; Component Map-Making (Chapter 6) instead includes a mixing matrix reconstruction into the map-making process, reconstructing astrophysical components and maps jointly. Between its Parametric and Blind variants, the latter proved necessary whenever the foreground spectral behaviour cannot be trusted a priori, in particular in the presence of frequency-decorrelated dust, at the cost of a larger parameter space and a substantially higher computational burden. Chapter 7 closed the loop from reconstructed maps to cosmological parameters through a cross-spectrum analysis, and in the process uncovered a practical but important effect: the number of noise realisations needed to obtain

an unbiased estimate of the tensor-to-scalar ratio grows sharply with the number of frequency channels included in the fit.

None of this would have been usable without the simulation and reconstruction software itself. A large share of this thesis, described in Chapter 8, went into consolidating *qubicsoft*: modernising its build system and its dependencies to restore compatibility with recent Python versions, although production simulations still rely on Python 3.10 pending further validation, and improving its documentation and testing. Additionally, a large refactoring of its architecture has been performed, to reach a more structured and general version. Chapter 9 used this consolidated pipeline to push the realism of the simulations further, in particular by studying the convergence of the simulated data with respect to the number of integrated sub-bands and the maximum diffraction order. An unexpected lack of convergence, eventually traced back to a pixelisation effect after several months of investigation, illustrates how demanding this realism requirement can be; the chapter also characterises reconstruction hyperparameters, such as the number of inner solver iterations, that affect the stability of the reconstruction. Not everything could be completed: the reconstruction of a spatially varying spectral index still diverges in the presence of noise for reasons that remain to be understood, and the reconstruction of the CO emission line, while demonstrated in principle, is not yet part of the routine pipeline. Finally, Chapter 10 explored an alternative to this forward-modelling philosophy altogether, using physics-guided neural networks to approximate the inverse of the instrumental model, with uncertainty quantification on the underlying physical parameters obtained through variational inference. This inverse approach is still at an early stage, but it already covers both the FMM and CMM formulations and offers a route to reconstruction that does not require repeated calls to the forward operator.

Atmospheric mitigation. Chapter 11 turned to a contaminant specific to ground-based experiments and left untreated by the rest of the thesis: the atmosphere. The starting observation was that the multi-peaked synthesized beam of a bolometric interferometer preserves atmospheric correlations over longer timescales than the narrow beam of a classical imager, which, combined with spectral imaging, makes it possible to treat the atmosphere as an additional component evolving in its own local, wind-advected domain rather than as noise to be filtered out. Extending the Blind CMM formalism to this problem, this thesis demonstrated as a proof of concept that the CMB, the atmosphere, the mixing matrix, and the wind vector can be reconstructed jointly. This result is, to my knowledge, the first application of map-making algorithm to mitigate atmospheric contamination, though it remains constrained by the computational cost and slow convergence of the alternating reconstruction scheme, which motivated the wind-free Successive Map-Making approach explored as a faster, if more approximate, alternative.

Perspectives. Several directions raised throughout this thesis point beyond it. The single most recurring one is algorithmic: both Component Map-Making and its extension to the atmosphere are currently limited by an alternating fitting scheme that converges slowly and scales poorly. Replacing it with a non-linear solver able to reconstruct all parameters simultaneously, or with a Gibbs-sampling scheme in the spirit of

Commander, would likely benefit every reconstruction method presented here at once, and is, in my view, the highest-impact development for the continuation of this work. Related software efforts are already underway, in particular the JAX-based rewrite of PyOperators and PySimulators led by Pierre Chanial, which should substantially improve the computational time that currently constrains both CMM and its atmospheric extension. On the instrumental and calibration side, the self-calibration uncertainty budget derived from GPS noise should be revisited with real, on-site data once available, and the reconstruction of a spatially varying spectral index needs to be understood before it can be relied upon for realistic sky models. For the atmosphere specifically, incorporating external measurements, whether from meteorological stations as priors or, more promisingly, from a dedicated LIDAR system, would relax the reconstruction's current dependence on the data alone. Finally, the neural-network-based inverse approach and the standard forward-modelling pipelines developed here are not in competition: bringing NNMM to the same level of maturity as FMM and CMM, including a dedicated sensitivity forecast, would make it possible to compare the two philosophies on equal footing.

Taken together, the work presented in this manuscript has moved QUBIC's data analysis from separate, single-purpose pieces towards a more unified pipeline, in which frequency map-making, component separation, neural-network-based inversion, and now atmospheric mitigation, are all expressed within the same spectral-imaging framework. Much of this pipeline remains to be optimised, and some of it remains to be finished, but the results obtained so far support the central premise of this thesis: that the specific capabilities of bolometric interferometry, self-calibration and spectral imaging in particular, are assets not only for controlling classical systematic effects, but also for tackling contaminants, like the atmosphere, that ground-based CMB experiments have traditionally had to filter out rather than model.